



Department of Information Technology

Academic Year 2024 - 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / **Equivalence of Finite Automata and Regular Expressions**
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen:** One Minute Paper
- **Justification:**

In unit II, the concept of equivalence between **Finite Automata and Regular Expressions** is foundational in automata theory and forms the basis for understanding **regular languages**. Despite being conceptually simple, students often find it abstract due to the transition from Finite Automata (FA) to Regular Expression (RE). This activity helps:

- Emphasize the conversion techniques between FA and RE.
 - Encourage critical thinking and retention through summarization.
 - Clarify doubts that determine higher-level understanding.
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- **Time Allotted for the Activity:** 5 Minutes
 - **Details of the Implementation:**
 - In the previous days the instructor explained the topic Equivalence of Finite Automata and Regular Expressions.
 - The time allotted for the topic is 5 Minutes. The students during the activity as shown in Figure 1
 - The students were asked to study the concepts of FA to RE and RE to FA
 - The next day, the students were asked to write down their doubts on a sheet of paper.
 - The papers were collected and corrected as shown in Figure 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	2	3	1	2	1	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

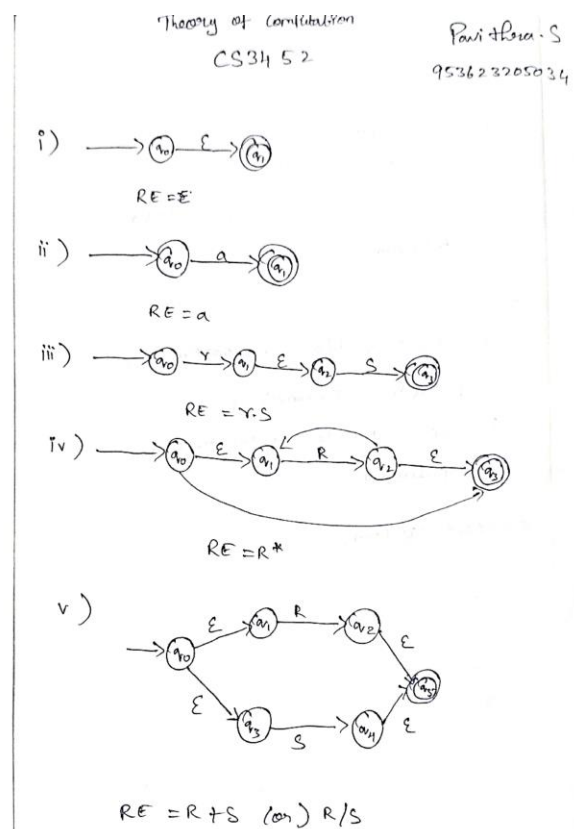
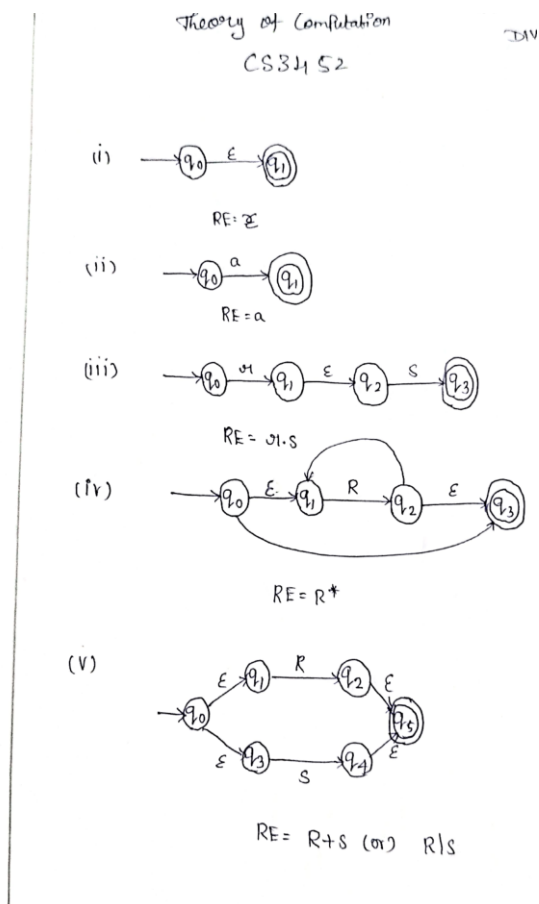
- **PO / PSO mapped:**

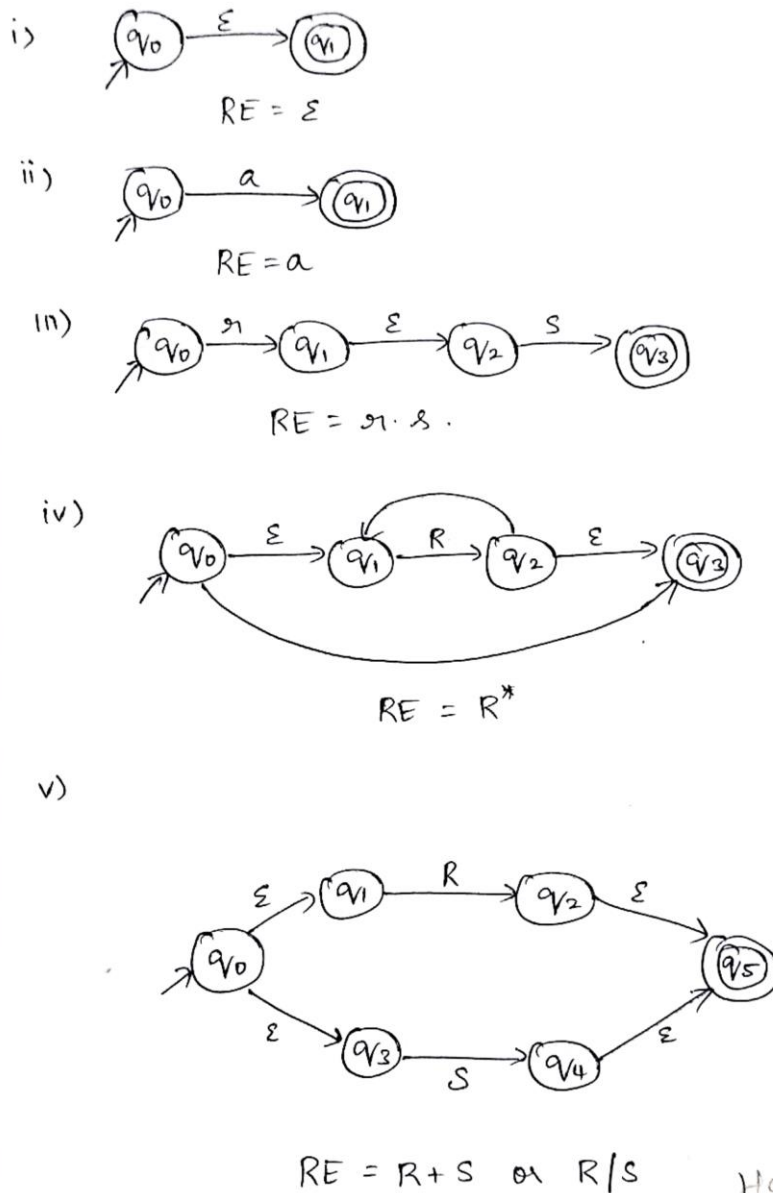
Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	2	3	1	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of equivalence of Finite Automata and Regular Expressions.	The students will be able to identify the need for Equivalence of Finite Automata and Regular Expressions in the context of complex engineering problems	Able to design solutions for any applications using equivalence of Finite Automata and Regular Expressions.	Able to analyze and apply Equivalence of Finite Automata and Regular Expressions. concept	Able to follow ethics for solving the problem	Able to implement the problem individually	Able to implement Equivalence of Finite Automata and Regular Expressions.	Ability to recognize the need for Equivalence of Finite Automata and Regular Expressions in real world applications	Exhibit knowledge in equivalence of Finite Automata and Regular Expressions for application development	Exhibit knowledge in Equivalence of Finite Automata and Regular Expressions of IoT, Cloud Computing and Data Analytics to provide IT solutions	Exhibit knowledge in Equivalence of Finite Automata and Regular Expressions in software projects to deliver a quality product.

- Images / Screenshot of the practice:



Figure 1: One Minute Paper Activity





Hari Eswar

Figure 2: Student Response Ms. J. S Divya Sri, S. Pavithra and A. Poornima

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students expressed that they could easily recall the concepts of Equivalence of Finite Automata and Regular expression, the conversion from RE to FA, FA to RE.
- They shared that the activity reinforced their understanding of the previous lesson and supported the development of independent learning skills.

- The activity also helped students grasp the conversion processes between FA and RE more effectively.

❖ ***Benefit of the practice:***

- Encouraged active reflection and reinforced the key learning point.
- Enabled identification of specific areas anywhere students need support.
- Fostered self-assessment skills and engagement in higher-order Some questions were difficult for students to understand the conversion.
- I felt the slow learners were not understood clearly. I explained separately for those students, then they understood clearly.

References:

- ❖ <https://ohioline.osu.edu/>
- ❖ <https://www.nureva.com/blog/education/15-active-learning-activities-to-energize-your-next-college-class>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member

HOD